

Pathological and Toxicological Findings in Glyphosate-Surfactant Herbicide Fatality

A Case Report

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Abstract: Glyphosate herbicide is promoted by the manufacturer as having no risks to human health, with acute toxicity being very low in normal use. In Thailand, however, poisoning from glyphosate agricultural herbicides has been increasing. A case of rapid lethal intoxication from glyphosate-surfactant herbicide involved a 37-year-old woman, who deliberately ingested approximately 500 mL of concentrated Roundup formulation (41% glyphosate as the isopropylamine salt and 15% polyoxyethylene amine; Monsanto Company). The postmortem examination revealed that the stomach contained 550 mL of yellow fluid. The gastric mucosa of anterior fundus revealed hemorrhage and the small intestines had marked dilatation and thin walls. We used the high-performance liquid chromatography method for determination of serum and gastric content levels of glyphosate. The glyphosate levels of serum and gastric content were 3.05 and 59.72 mg/mL, respectively. Toxic effects of polyoxyethylene amine and Roundup were caused by their ability to erode tissues including mucous membranes and linings of the gastrointestinal and respiratory tracts. A mild degree of pulmonary congestion and edema was observed in both lungs. We proposed that the characteristic picture of microvesicular steatosis of the hepatocytes, seen predominantly in centrilobular zones of the liver, resembled drug-induced hepatic toxicity or secondary hypoxic stress.

Key Words: glyphosate, acute toxicity, serum level, hepatic toxicity

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Today, Roundup and other glyphosate agricultural herbicides are among the world's most widely used herbicides.¹ Glyphosate, *N*-(phosphonomethyl)glycine, is an organophosphate herbicide, usually formulated with the organic base, isopropylamine, to yield a more water-soluble salt. This salt, combined with water and a surfactant to improve performance in the field, comprises the principal glyphosate-surfactant herbicide (GlySH) sold worldwide under the Roundup family of brand names.^{2,3} Roundup contains surfactant polyoxyethylene amine (POEA). Polyoxyethylene amine is a nonionic surfactant that promotes the penetration of glyphosate into the plant cuticle, thereby improving its effectiveness.⁴

Glyphosate is a systemic and nonselective herbicide used to kill grass, broadleaved, and sedge species. The toxic effect on plants is caused by competitive inhibition of the enzyme, 5-enolpyruvyl-

shikimic-3-phosphate synthase, used by plants to make certain important aromatic amino acids. Without these amino acids, the plant cannot make the proteins required for various life processes, resulting in the death of the plant. The enzyme, 5-enolpyruvyl-shikimic-3-phosphate synthase, exists in plants and microorganisms, but not in animals.^{2,3} It was therefore considered for selective glyphosate toxicity to plant species, with lower toxicity to humans and animals than other commonly used herbicides. Glyphosate is not a cholinesterase inhibitor and does not affect the nervous system in the same way as organophosphate insecticides. The primary degradation product of glyphosate in plants, soil, and water is aminomethylphosphonic acid, which has a chemical structure very similar to that of glyphosate. Aminomethylphosphonic acid was considered to be of no greater toxicological concern than its parent compound,³ and residual glyphosate also biodegrades very satisfactorily in soil environments.⁵ Both glyphosate and aminomethylphosphonic acid are classified by the US Environmental Protection Agency as being in the least toxic category (IV), with acute toxicity being very low in normal use.⁶ Glyphosate-surfactant herbicide is promoted by the manufacturer as having no risks to human health and the environment when used according to label directions.

Suicide attempts with agricultural chemicals are common in Thailand, but GlySH is not commonly used. However, GlySH is widely available as a retail purchase, with the potential for serious and lethal toxicity when taken for suicide. Many cases of acute intoxication with GlySH have been reported in many countries. However, relatively little literature exists regarding the toxicology and pathology of GlySH in humans. We report a case of GlySH toxicity and death. The pathological findings, together with toxicological analysis, are disclosed in this case.

CASE REPORT

Case History

A 37-year-old woman was found dead in her bedroom. Police investigators found a bottle of Roundup, labeled as GlySH, nearby the corpse and reported that the decedent had drunk approximately 500 mL from it. The body was transferred to the Department of Forensic Medicine at Chiang Mai University for a medicolegal autopsy. The autopsy was performed 10 hours after the deceased was found.

Autopsy Findings

The deceased was a 37-year-old Asian woman. There was no violent wound found on the body. Rigor mortis and livor mortis were noticed. Yellow fluids emerged from the nose. Central and peripheral cyanosis was observed. The scalp was congested, with no wound. The brain weighed 1100 g and revealed marked congestion. The heart weighed 290 g with no coronary artery occlusion. Both lungs weighed 680 g and showed pulmonary congestion

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and edema. The liver, pancreas, and spleen weighed 1500, 110, and 70 g, respectively, and revealed moderate congestion. Both kidneys weighed 274 g together. A yellow fluid content of 550 mL was found in the stomach. The gastric mucosa of the anterior fundus revealed hemorrhage (Fig. 1). The small intestines had marked dilatation and thin walls. Air-fluid levels were observed in multiple loops of intestine (Fig. 2).

Microscopic Findings

Microscopic examination of tissue sections of the brain revealed congestion and edema. Sections of the heart displayed mild hypertrophy of myocardial fibers and vascular congestion. A mild degree of pulmonary congestion and edema was noted in both lungs (Fig. 3). Sections of the liver revealed marked sinusoidal congestion and mild fatty change of microvesicular and macrovesicular types at centrilobular areas (Fig. 4). The kidneys revealed vascular congestion, with proteinaceous materials in glomerular tufts and lumens of renal tubules. Focal areas of chronic pyelonephritis were observed in the left kidney. Both adrenal glands showed medullary congestion. There was no significant microscopic change in the spleen or pancreas. Sections of the esophagus displayed multiple areas of microscopic erosion of the squamous mucosa. The gastric mucosa revealed partial degeneration of epithelial lining cells, which could be either partly coagulative necrosis or partial autolytic change.

TOXICOLOGICAL ANALYSIS

Specimen Collections

Heart blood, gastric content, and yellow fluid in the Roundup bottle, found at the death scene, were submitted for toxicological investigation. The analysis of glyphosate was conducted using both thin-layer chromatography and high-performance liquid chromatography. Femoral blood and vitreous fluids were taken and submitted to the laboratory for alcohol analysis.

TOXICOLOGICAL RESULTS

Using thin-layer chromatography, glyphosate and surfactants of Roundup were identified in the serum, gastric content, and the fluid in the Roundup bottle. By using the high-performance liquid chromatography technique, the glyphosate levels in the serum and gastric content were 3.08 and 59.72 mg/mL, respectively. Alcohol was not found in either femoral blood or vitreous fluids.



FIGURE 1. Photograph of the stomach revealing gastric mucosal hemorrhage.



FIGURE 2. Photograph of small intestines demonstrating bowel dilatation with marked air-fluid levels.

Cause of Death

The cause of death in this victim was GlySH intoxication. The manner of death was suicide.

DISCUSSION

Glyphosate (CAS No. 1071-83-6) is a postemergent, systemic, and nonselective herbicide that is used in both agricultural and nonagricultural areas all over the world. The major formulation, GlySH, is one of the most broadly used herbicides today.¹ The acute toxicity of glyphosate itself is very low. The acute oral median lethal dose of glyphosate in rats is greater than 4320 mg/kg of body weight. The acute toxicity of the surfactant, POEA, is somewhat higher than that for GlySH formulation. The oral median lethal dose value in rats has been reported as 1200 mg/kg.^{3,5} Glyphosate-surfactant herbicide is placed in the US Environmental Protection Agency's least toxic category (IV) for acute oral, dermal, and inhalation toxicity. Previous studies support that GlySH is minimally toxic in the setting of small volume, accidental, or occupational exposure (including dermal, ocular, inhalational, and oral). No deaths have been reported after accidental ingestion of GlySH.^{7,8}

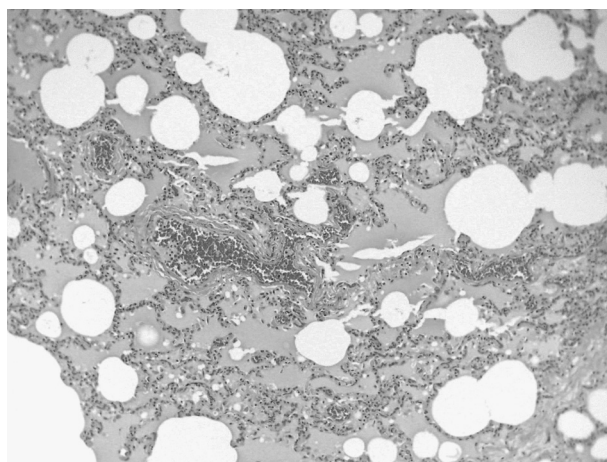


FIGURE 3. Photomicrograph of the lung section showing pulmonary vascular congestion and edematous fluids in alveolar spaces (hematoxylin-eosin, $\times 100$).

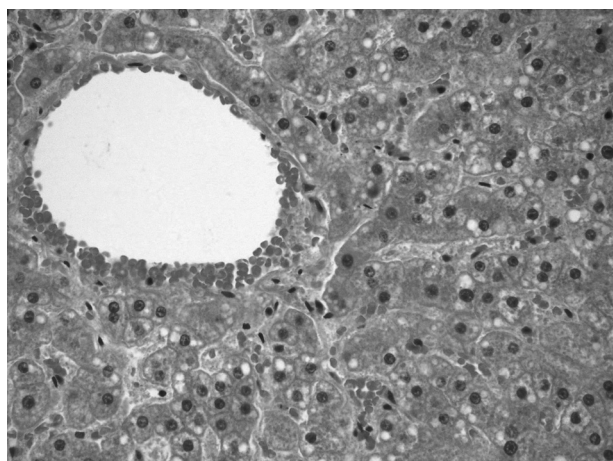


FIGURE 4. Photomicrograph of the liver section showing sinusoidal congestion and predominantly microvesicular fat deposition in the hepatocytes of a centrilobular area (hematoxylin-eosin, $\times 400$).

Although GlySH poisoning is uncommon, suicidal ingestion of GlySH can result in serious and fatal sequelae. The decedent in this case deliberately ingested approximately 500 mL of concentrated Roundup formulation (41% glyphosate as the isopropylamine salt and 15% POEA). Ingestion of small amounts (<150 mL) of glyphosate herbicide usually causes only mild symptoms.⁷ However, when large volumes of GlySH are ingested intentionally, it can generate potentially fatal symptoms that are refractory to treatment. It was estimated that intentional ingestion of large amounts (>200 mL) of concentrated GlySH in suicide attempts is highly associated with fatal outcome.^{8,9} Deaths after ingestion of GlySH are caused by severe hypotension, respiratory distress, renal failure, metabolic acidosis, and hyperkalemia. In such cases, death usually occurs within a few days of ingestion.^{8,10}

Orally administered glyphosate is poorly absorbed from the alimentary tract despite the prevailing acidic conditions.³ Toxic effects of POEA and Roundup are caused by their ability to erode tissues including mucous membranes and linings of the gastrointestinal and respiratory tracts. Common effects after accidental or intentional ingestion of Roundup include mucosal erosion or ulceration; resulting in sore throat, dysphagia, and gastrointestinal hemorrhage.^{7,8,10} Erosion of the esophageal mucosa and partly coagulative necrosis were also demonstrated in our reported case. This suggested that in a case of GlySH poisoning, gastric intubation should be conducted with caution to prevent gastroesophageal perforation. Blood level of glyphosate was also determined in this case of intentional massive ingestion. The serum glyphosate level in our reported case was 3.05 mg/mL, whereas in the asymptomatic and mild clinical manifestation reported by Talbot et al,⁷ the level was up to 0.1 mg/mL and in severe cases, greater than 1 mg/mL. In a case of nonfatal GlySH poisoning, the level reported was 0.022 mg/mL.¹¹ The pathological findings in our case showed that the gastric mucosa was eroded, and this may have allowed more glyphosate and surfactants to be absorbed into the circulation.

The relatively small amounts of absorbed glyphosate were rapidly excreted in urine, almost exclusively as unchanged parent material. The tissue distribution of glyphosate was reported in an animal model. Tissue retention times were relatively short, and most of body burden was unmetabolized parent glyphosate. Significant glyphosate retention was detected in the small intestine, colon, kidneys, spleen, fat, and liver. Maximum concentrations in the small intestine (primarily associated with cells rather than

contents) and blood were observed 2 hours after oral glyphosate administration, whereas peak levels in other organs occurred 6.3 hours after dosing.¹²

The commonly used component of glyphosate products, POEA, is 2 to 3 times more toxic than glyphosate; and the formulated product containing POEA, such as Roundup, may be even more toxic.^{13,14} Purposed mechanisms of serious systemic sequelae or death from large amounts of GlySH ingestion were POEA- or glyphosate-mediated direct cardiotoxicity and pulmonary damage, or death through refractory circulatory failure. Renal and hepatic impairment were frequent and usually reflected reduced organ perfusion. Respiratory distress, pulmonary edema, impaired consciousness, shock, arrhythmias, renal failure, metabolic acidosis, and hyperkalemia were present in severe cases.^{10,15} We demonstrated lung pathology as a mild to moderate degree of pulmonary edema in our case (Fig. 3). Microscopic pathology of myocardial fibers could not be demonstrated. The nature of the clinical symptoms observed in cases of suicide suggests that hypovolemic shock is the cause of death.^{9,15} It has been suggested that the acute toxicity of Roundup is likely caused by the surfactant because similar responses have been observed in cases involving ingestion of other surface active agents. The hypovolemic shock is caused by the cardiac depressant effect of very high doses of the surfactant.⁵

Glyphosate was shown to decrease the hepatic function of cytochrome P-450 and monooxygenase activities.¹⁶ The P-450 enzyme system is one of the main body systems for detoxifying harmful chemicals. When this enzyme function becomes impaired by chemicals supposed to be detoxified, the effects of a given chemical on the body increase dramatically. Glyphosate exposure and metabolism in the liver can lead to excessive production of malondialdehyde and oxidative stress through unregulated generation of reactive oxygen species such as superoxide anion, hydrogen peroxide, hydroxyl radical, peroxyl radicals, and singlet oxygen. Excessive reactive oxygen species can compromise cellular integrity through oxidative damage of lipids, proteins, or DNA.¹⁷ We proposed that the characteristic picture of microvesicular steatosis of the hepatocytes, seen predominantly in centrilobular zones of the liver (Fig. 4), resembled drug-induced hepatic toxicity or secondary hypoxic stress. The intracytoplasmic accumulation of small fat vacuoles in hepatocytes could be an acquired defect in the β -oxidation of fatty acids.¹⁸

In conclusion, ingesting high dosage levels of GlySH is acutely toxic to humans and can be fatal. We demonstrated a GlySH (Roundup) fatality case. This study measured the lethal blood level of glyphosate, which was higher than levels reported previously. Toxic-related effects were observed in the gastrointestinal tract, lungs, and liver. Knowing these pathological changes creates more understanding of the toxicity mechanism as well as the limitation of treatment.

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